

## MOORED BUOY 21004, Japan

WMO: 992280

Lat: 29.000N

Lon: 135.000E

Elev: 1

StdP: 101.31

Time Zone: 9.00 (E09)

Period: 90-00

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 11.8       | 12.7 | 2.1                           | 4.4 | 13.3 | 3.2 | 4.8 | 13.7 | 19.6                  | 15.0 | 18.2 | 14.4 | 9.3                   | 340  | 0.556 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 1.5                    | 30.0            | 26.5 | 29.7 | 26.4 | 29.4 | 26.3 | 27.7                | 28.9 | 27.3 | 28.8 | 27.1 | 28.8 | 4.8                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.2                            | 23.0 | 28.6 | 26.9 | 22.6 | 28.5 | 26.6 | 22.1 | 28.4 | 88.5          | 29.0 | 86.8 | 28.9 | 85.4 | 28.7 | 30.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |
| 20.9              | 16.2 | 13.7 | DB                         | 10.3 | 30.9               | 0.7 | 1.7                                                | 9.8 | 32.2       | 9.3 | 33.1       | 8.9 | 34.1       | 8.4 | 35.3 |
|                   |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A | N/A  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 22.2  | 16.7  | 16.0  | 17.6  | 19.5  | 21.8  | 24.8  | 27.7  | 28.2  | 27.5  | 25.0  | 21.9  | 18.6  | (6)  |
| (7)  |                                                                                  | DBStd       | 4.61  | 2.18  | 2.17  | 2.02  | 1.94  | 1.66  | 1.66  | 1.22  | 1.02  | 1.20  | 1.69  | 1.78  | 2.14  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |
| (9)  |                                                                                  | HDD18.3     | 201   | 60    | 68    | 38    | 11    | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 23    | (9)  |
| (10) |                                                                                  | CDD10.0     | 4437  | 207   | 169   | 236   | 286   | 366   | 445   | 548   | 564   | 526   | 466   | 358   | 267   | (10) |
| (11) |                                                                                  | CDD18.3     | 1597  | 9     | 4     | 16    | 47    | 108   | 195   | 290   | 306   | 276   | 207   | 108   | 32    | (11) |
| (12) |                                                                                  | CDH23.3     | 12907 | 0     | 0     | 1     | 11    | 156   | 1222  | 3252  | 3620  | 3053  | 1420  | 169   | 3     | (12) |
| (13) | CDH26.7                                                                          | 3127        | 0     | 0     | 0     | 1     | 1     | 84    | 918   | 1193  | 813   | 117   | 1     | 0     | (13)  |      |
| (14) | Wind                                                                             | WSAvg       | 6.7   | 7.5   | 7.7   | 6.7   | 6.1   | 5.7   | 6.2   | 6.0   | 6.2   | 7.2   | 7.4   | 7.2   | 6.9   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1741  | 76    | 76    | 96    | 159   | 257   | 303   | 120   | 121   | 152   | 177   | 111   | 93    | (15) |
| (16) |                                                                                  | PrecMax     | 2550  | 202   | 201   | 197   | 348   | 509   | 630   | 289   | 313   | 359   | 359   | 224   | 208   | (16) |
| (17) |                                                                                  | PrecMin     | 1317  | 44    | 32    | 30    | 27    | 71    | 102   | 9     | 14    | 41    | 65    | 30    | 32    | (17) |
| (18) |                                                                                  | PrecStd     | 226   | 39    | 39    | 42    | 73    | 106   | 131   | 79    | 73    | 81    | 76    | 49    | 39    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 22.1  | 21.5  | 22.5  | 24.5  | 26.7  | 28.5  | 30.6  | 30.4  | 30.1  | 28.7  | 26.2  | 23.8  | (19) |
| (20) |                                                                                  |             | MCWB  | 19.7  | 19.5  | 20.8  | 23.6  | 25.4  | 26.2  | 26.6  | 26.6  | 27.1  | 25.4  | 23.6  | 21.4  | (20) |
| (21) |                                                                                  | 2%          | DB    | 21.2  | 20.6  | 21.7  | 23.3  | 26.0  | 28.0  | 30.0  | 30.0  | 29.7  | 28.2  | 25.2  | 23.1  | (21) |
| (22) |                                                                                  |             | MCWB  | 18.8  | 18.5  | 20.1  | 22.2  | 24.1  | 25.8  | 26.4  | 26.5  | 26.5  | 25.1  | 22.6  | 20.1  | (22) |
| (23) |                                                                                  | 5%          | DB    | 20.6  | 19.9  | 21.1  | 22.5  | 25.2  | 27.6  | 29.6  | 29.7  | 29.4  | 27.8  | 24.8  | 22.5  | (23) |
| (24) |                                                                                  |             | MCWB  | 17.9  | 17.4  | 19.6  | 21.4  | 23.4  | 25.6  | 26.2  | 26.4  | 26.3  | 24.9  | 21.8  | 19.2  | (24) |
| (25) |                                                                                  | 10%         | DB    | 20.0  | 19.2  | 20.5  | 22.0  | 24.0  | 27.2  | 29.2  | 29.4  | 29.1  | 27.4  | 24.3  | 21.8  | (25) |
| (26) |                                                                                  |             | MCWB  | 16.8  | 16.3  | 18.9  | 20.8  | 22.3  | 25.3  | 26.1  | 26.3  | 26.2  | 24.6  | 21.2  | 18.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 20.6  | 20.4  | 21.1  | 22.6  | 24.0  | 27.1  | 28.0  | 28.3  | 27.9  | 26.7  | 24.6  | 22.5  | (27) |
| (28) |                                                                                  |             | MADB  | 21.7  | 21.3  | 21.7  | 22.7  | 24.3  | 27.7  | 29.1  | 29.1  | 29.1  | 27.8  | 25.4  | 23.4  | (28) |
| (29) |                                                                                  | 2%          | WB    | 19.5  | 19.0  | 20.4  | 21.8  | 23.3  | 26.6  | 27.6  | 27.7  | 27.3  | 26.0  | 23.6  | 21.0  | (29) |
| (30) |                                                                                  |             | MADB  | 21.0  | 20.2  | 21.0  | 22.1  | 23.7  | 27.3  | 29.0  | 28.9  | 29.0  | 27.6  | 24.8  | 22.7  | (30) |
| (31) |                                                                                  | 5%          | WB    | 18.4  | 17.9  | 19.9  | 21.3  | 22.8  | 26.2  | 27.2  | 27.4  | 27.0  | 25.6  | 22.7  | 19.8  | (31) |
| (32) |                                                                                  |             | MADB  | 20.3  | 19.6  | 20.6  | 21.7  | 23.4  | 27.0  | 28.8  | 28.9  | 28.9  | 27.3  | 24.3  | 22.0  | (32) |
| (33) |                                                                                  | 10%         | WB    | 17.2  | 16.7  | 19.2  | 20.8  | 22.3  | 25.8  | 26.9  | 27.1  | 26.7  | 25.2  | 21.9  | 18.8  | (33) |
| (34) |                                                                                  |             | MADB  | 19.7  | 18.9  | 20.2  | 21.4  | 23.0  | 26.6  | 28.7  | 28.8  | 28.7  | 27.1  | 23.9  | 21.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 2.5   | 2.5   | 2.4   | 2.0   | 1.8   | 1.6   | 1.7   | 1.5   | 1.8   | 1.9   | 2.1   | 2.3   | (35) |
| (36) |                                                                                  |             | MCDBR | 3.3   | 2.9   | 2.8   | 2.3   | 1.9   | 1.4   | 2.0   | 1.6   | 1.8   | 1.9   | 2.5   | 2.6   | (36) |
| (37) |                                                                                  |             | MCWBR | 4.6   | 4.4   | 3.4   | 2.5   | 3.2   | 1.1   | 1.2   | 1.1   | 1.4   | 1.7   | 2.9   | 3.5   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 3.3   | 3.0   | 2.5   | 2.1   | 2.0   | 1.6   | 1.9   | 1.3   | 1.8   | 1.7   | 2.4   | 2.5   | (38) |
| (39) |                                                                                  |             | MCWBR | 4.9   | 4.9   | 3.4   | 2.8   | 2.3   | 1.5   | 1.6   | 1.2   | 1.5   | 1.9   | 3.4   | 4.0   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |       | 0.427 | 0.455 | 0.505 | 0.525 | 0.498 | 0.463 | 0.465 | 0.466 | 0.459 | 0.459 | 0.433 | 0.426 | (40) |
| (41) |                                                                                  | taud        |       | 2.196 | 2.136 | 2.009 | 2.008 | 2.179 | 2.365 | 2.357 | 2.355 | 2.351 | 2.286 | 2.295 | 2.206 | (41) |
| (42) |                                                                                  | Ebn at Noon |       | 792   | 804   | 789   | 787   | 808   | 831   | 829   | 826   | 822   | 793   | 784   | 771   | (42) |
| (43) |                                                                                  | Edh at Noon |       | 124   | 143   | 172   | 177   | 150   | 124   | 125   | 124   | 121   | 122   | 112   | 118   | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      |       | 2.60  | 3.22  | 4.22  | 4.90  | 5.03  | 5.15  | 6.55  | 6.29  | 5.34  | 4.02  | 3.03  | 2.41  | (44) |
| (45) |                                                                                  | RadStd      |       | 0.24  | 0.32  | 0.33  | 0.47  | 0.62  | 0.62  | 0.36  | 0.25  | 0.32  | 0.27  | 0.25  | 0.20  | (45) |

## Historical Trends

|      |                        | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |
|------|------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A |
| (47) | Regional (0 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A |

Nomenclature: See separate page